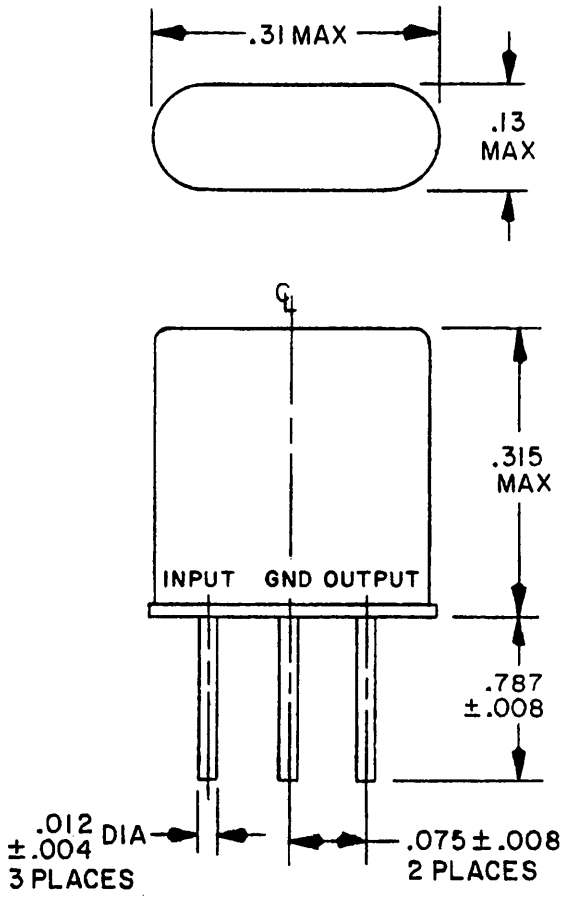
MILITARY SPECIFICATION SHEET

FILTERS, BAND PASS, .315 INCH CAN STYLE

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the filters described herein shall consist of this Specification Sheet and the latest issue of MIL-F-18327.

INCHES	MM
.004	0.10
.008	0.20
.012	0.30
.075	1.91
.13	3.3
.31	7.9
.315	8.00
.787	19.99



NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only.
3. These filters are electrically symmetrical and may be installed in reverse position without changing the performance characteristics.

FIGURE 1. Case and mounting dimensions.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.

Case: Metal.

Terminals: Solderable pin.

Weight: .02 ounces maximum.

Terminal strength: Method 211, MIL-STD-202. Test condition A (pull), 9 newtons.

Solderability: Method 208, MIL-STD-202. All terminations of each filter to be tested.

Solder dip dwell time: Method 208, MIL-STD-202.

Examination of terminations: Method 208, MIL-STD-202.

Dielectric withstanding voltage:

Terminals to case:

At atmospheric pressure: Method 301, MIL-STD-202. 100 V rms between each terminal (except ground) and case.

At reduced pressure: In accordance with MIL-F-18327.

Input to output: Method 301, MIL-STD-202. The following details shall apply:

Magnitude and nature of test voltage: 50 V dc.

Duration of application of test voltage: 5 ±1 second. The test voltage shall be raised from 0 to the specified value within 1 second maximum.

Points of application of test voltage: Between input and output terminals. Test shall be repeated with reverse voltage polarity.

Limiting of surge current: 30 to 50 mA.

Examinations after test: Filters shall be examined for evidence of breakdown and damage.

Insulation resistance: Method 302, test condition B, MIL-STD-202. Not less than 10,000 megohms when measured between each terminal (except ground) and case.

Stability at temperature extremes:

1 hour at -20°C, +0°C, -3°C.

1 hour at 70°C, +3°C, -0°C.

Measurements at 25°C after test:

Dielectric withstanding voltage - At atmospheric pressure.

Electrical characteristics - As specified in table I.

Life: Method 108, test condition F, MIL-STD-202.

Test temperature 70°C, +10°C, -5°C.

Signal input level - See table I.

Measurements at 25°C after test:

Dielectric withstanding voltage - At atmospheric pressure.

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Insulation resistance - Not less than 1,000 megohms between each terminal (except ground) and case.

Electrical characteristics - As specified in table I.

Temperature rise: Not applicable.

Vibration, high frequency: Method 204, test condition B, MIL-STD-202.

Electrical load not applicable.

Measurements at 25°C after test:

Dielectric withstanding voltage - At atmospheric pressure.

Electrical characteristics - As specified in table I.

Shock (specified pulse): Method 213, test condition I, MIL-STD-202.

Measurements at 25°C after test:

Dielectric withstanding voltage - At atmospheric pressure.

Electrical characteristics - As specified in table I.

Thermal shock: Method 107, test condition A, MIL-STD-202, except that the temperature at step 1 shall be -20°C and at step 3 shall be 70°C.

Exposure time - 30 minutes.

Measurements at 25°C after test:

Dielectric withstanding voltage - At atmospheric pressure.

Insulation resistance - Not less than 10,000 megohms between each terminal (except ground) and case.

Immersion: Method 104, test condition A, MIL-STD-202. After final cycle, filters shall be washed under running tap water and dried. After the drying period, the filters shall be subjected to the following:

Measurements at 25°C.

Dielectric withstanding voltage - At atmospheric pressure.

Insulation resistance - Not less than 10,000 megohms between each terminal (except ground) and case.

Moisture resistance: Method 106, MIL-STD-202.

Measurements at 25°C after test:

Dielectric withstanding voltage - At atmospheric pressure.

Insulation resistance - Not less than 10,000 megohms between each terminal (except ground) and case.

Flammability: Not applicable.

Part number: M18327/70- (dash number from table I).

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TABLE I. Electrical characteristics. 1/ 2/

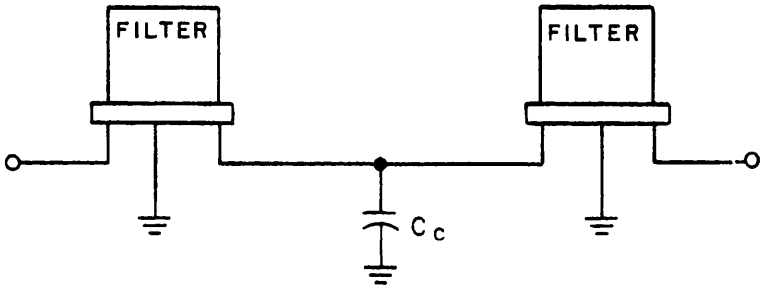
Dash number	Impedance (ohms) Source and load	Capacitance -0, +100 percent (pF)	Reference frequency (MHz)	Insertion loss (dB)	Frequency range (kHz)	Discrimination (dB) Min	Discrimination (dB) Max	Rated voltage (mV rms)
001	850	6.0	21.4	2.0	21,382	18		100
			"	"	21,396.25		3	"
			"	"	21,403.75		3	"
			"	"	21,418	18		"
002	910	5.0	"	"	21,378	18		"
			"	"	21,394		3	"
			"	"	21,406		3	"
			"	"	21,422	18		"
003	16 k	1.0	"	"	21,370	18		"
			"	"	21,392.5		3	"
			"	"	21,407.5		3	"
			"	"	21,430	18		"
004	"	"	"	"	21,366	18		"
			"	"	21,390		3	"
			"	"	21,410		3	"
			"	"	21,434	18		"
005	2.3 k	-2.0 3/	"	"	21,350	18		"
			"	"	21,385		3	"
			"	"	21,415		3	"
			"	"	21,450	18		"
006	3.0 k	-1.0 3/	45	1.5	44,975	15		"
			"	"	44,990		3	"
			"	"	45,010		3	"
			"	"	45,025	15		"
007	2.5 k	"	70	"	69,975	13		"
			"	"	69,990		3	"
			"	"	70,010		3	"
			"	"	70,025	13		"
008	"	"	90	"	89,975	13		"
			"	"	89,990		3	"
			"	"	90,010		3	"
			"	"	90,025	13		"

1/ Ripple: 0.5 dB maximum throughout the 3 dB passband.
2/ VSWR: 4.5 maximum throughout the 3 dB passband.
1.5 maximum at the reference frequency.
3/ A negative value for capacitance indicates inductive reactance.

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APPLICATION INFORMATION:

With two crystal filters connected in tandem as shown on figure 2 with the coupling capacitor as specified in table II, the electrical characteristics shall be as specified in table II.



NOTES:
1. The two filters are the same dash number specified in table II.
2. The C_C is called the coupling capacitor as specified in table II.
3. Ripple: 1.0 dB maximum throughout the 3 dB passband.
4. VSWR: 4.5 maximum throughout the 3 dB passband.
   1.3 maximum at the reference frequency.

FIGURE 2. Tandom crystal filters.

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TABLE II. Tandem electrical characteristics. 1/ 2/

Coupling capacitor (pF) (MIL-C-20) part number	Dash number	Impedance (ohms) Source and load	Capacitance -0, +100 percent (pF)	Reference frequency (MHz)	Insertion loss (dB)	Frequency range (kHz)	Discrimination (dB) Min	Discrimination (dB) Max	Rated voltage (mV rms)
15 (CCR 05CG 150JP)	001	850	5.0	21.4	2.5	21,386	40		100
				"	"	21,396.25		3	"
				"	"	21,403.75		3	"
				"	"	21,414	40		"
12 (CCR 05CG 120JP)	002	910	5.0	"	"	21,380	40		"
				"	"	21,394		3	"
				"	"	21,406		3	"
				"	"	21,420	40		"
6.8 (CCR 05CH 6R8DP)	003	16 k	1.0	"	"	21,375	40		"
				"	"	21,392.5		3	"
				"	"	21,407.5		3	"
				"	"	21,425	40		"
3.9 (CCR 05CH 3R9CP)	004	"	"	"	"	21,366	40		"
				"	"	21,390		3	"
				"	"	21,410		3	"
				"	"	21,434	40		"
1.5 (CCR 05CH 1R5CP)	005	2.3 k	-2.0 3/	"	"	21,350	40		"
				"	"	21,385		3	"
			-	"	"	21,415		3	"
				"	"	21,450	40		"
-1.5	006	3.0 k	-1.0 3/	45	3.0	44,975	25		"
				"	"	44,990		3	"
				"	"	45,010		3	"
				"	"	45,025	25		"
-1	007	2.5 k	"	70	"	69,975	25		"
				"	"	69,990		3	"
				"	"	70,010		3	"
				"	"	70,025	25		"
-1	008	"	"	90	"	89,975	25		"
				"	"	89,990		3	"
				"	"	90,010		3	"
				"	"	90,025	25		"

1/ Ripple: 1.0 dB maximum throughout the 3 dB passband.
2/ VSWR: 4.5 maximum throughout the 3 dB passband.
1.3 maximum at the reference frequency.
3/ A negative value for capacitance indicates inductive reactance.

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QUALITY ASSURANCE PROVISIONS:

Extent of Qualification:

Qualification testing and approval to M18327/70-001 shall be sufficient to grant qualification approval to M18327/70-001 thru M18327/70-008.

Qualification testing and approval to M18327/69-001, /70-001, /71-001, /72-001, /73-011, /74-001, /75-005, /76-024, and /77-001 shall be sufficient to grant qualification approval to MIL-F-18327/69 through MIL-F-18327/77, inclusive.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
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Navy - AS, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - ME
Navy - MC
Air Force - 19

Agent:
DLA - ES

Preparing activity:
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